

MONIKA SINGH

AI / ML Engineer | Software Developer | Information Technology

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PROFESSIONAL SUMMARY

Computer Engineering graduate currently working in IT at HDFC Bank, with a strong focus on AI/ML and software development. I enjoy building machine learning tools and full-stack applications, and have worked on projects across NLP, computer vision, and data analysis. Comfortable writing Python with common ML libraries, and familiar with everyday tools like Git, Docker, and cloud services.

PROFESSIONAL EXPERIENCE

Assistant Manager (IT) | HDFC Bank Ltd, India

Oct 2024 – Present

- Build and maintain internal tools that automate routine banking tasks, using Python and machine learning to reduce manual work.
- Work on data pipelines and reporting dashboards that help teams track key numbers and make decisions faster.
- Develop backend services and REST APIs, and use Docker and Git as part of the regular development workflow.
- Handle sensitive financial data carefully, following RBI rules and the bank's internal security policies.

Software Engineer Intern | Kotak Mahindra Bank, India

Jun 2023 – Aug 2023

- Helped build and test features for internal web applications using Python, JavaScript, and SQL.
- Wrote scripts to clean and process data, and got early hands-on experience with machine learning workflows.

PROJECTS

Enterprise Knowledge Assistant (RAG) | Python, LangChain, FAISS, OpenAI API, Flask

Feb 2025 – Apr 2025

- Built a question-answering tool that searches internal documents and returns relevant answers using embeddings and a vector database.
- Set up the document processing and retrieval steps, and served the model through a simple Flask API.

Customer Churn Prediction System | Python, scikit-learn, Pandas, Streamlit

Aug 2024 – Oct 2024

- Built a model to predict which customers were likely to leave, using feature engineering and tuning to improve the results.
- Checked performance with precision, recall, and AUC, and put the model behind a Streamlit app for quick predictions.

Real-Time Vision Detection System | Python, OpenCV, TensorFlow, CNN

Mar 2023 – May 2023

- Trained a CNN to detect and classify objects from a live webcam feed, using data augmentation and transfer learning.
- Tuned the pipeline so it runs smoothly in real time across different lighting and angles.

TECHNICAL SKILLS

Programming: Python, C++, Java, JavaScript, Go, SQL, C

AI / ML: TensorFlow, PyTorch, Keras, scikit-learn, OpenCV, NLP, Computer Vision, CNNs, Model Evaluation

GenAI / LLM: LangChain, RAG, Prompt Engineering, OpenAI / Gemini APIs, Vector DBs (FAISS, Chroma)

Data & Viz: Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Power BI

Web: Node.js, Express, React, HTML5, CSS3, REST APIs, Flask

Databases: MySQL, PostgreSQL, MongoDB

DevOps & Tools: Docker, Git, GitHub, Linux, AWS (S3, EC2), CI/CD, VS Code, Jupyter, Google Colab

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, System Design

EDUCATION

B.Tech, Computer Engineering | Delhi Technological University (DTU), New Delhi

2020 – 2024

Senior Secondary (Class XII), CBSE | Govt. Girls Senior Secondary School

2020

CERTIFICATIONS & ACHIEVEMENTS

- Studied machine learning and deep learning through online courses alongside my degree.
- Solved 300+ coding problems on LeetCode and HackerRank, and keep several AI/ML projects on GitHub.